

Representations, irreducibility and equivalence

[Back to Project mathematical routes](#)

Scope

Representations without a unit equation, nonzero irreducibility and unitary equivalence set the language of the theorem. The singleton condition names one representative of the unique nonzero irreducible class.

4 direct Cards + 0 reused prerequisites = 4 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key definition: [A representative of the unique irreducible class](#)
- Key definition: [Representations without a unit-preservation requirement](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

[0](#) [1](#) [2](#)

Level 0 (1 Card)

Level 0

Representations without a unit-preservation requirement

Names *-representations of a complex C^* -algebra on a Hilbert space when no unit equation is required.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1 (2 Cards)

Level 1

Unitary equivalence of two representations

Defines equivalence by one surjective isometry intertwining all algebra actions.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.UnitaryEquivalent`

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

[Open Declaration Card](#) [Read exact source](#)

Level 1

Nonzero irreducibility without a unit assumption

Requires nonzero action as well as the absence of proper closed reducing subspaces.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsIrreducible`

Immediate Card prerequisites: [Representations without a unit-preservation requirement](#)

Used by in this scope: [A representative of the unique irreducible class](#)

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Level 2 (1 Card)

Level 2

A representative of the unique irreducible class

Defines when one nonzero irreducible $*$ -representation represents the sole unitary-equivalence class of such representations.

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.IsSingletonIrreducibleModel`

Immediate Card prerequisites: [Nonzero irreducibility without a unit assumption](#) · [Unitary equivalence of two representations](#)

Used by in this scope: None in this selected scope

[Open Declaration Card](#) [Read exact source](#)